MCS 320 Project Two : Integration to Approximate Functions
due Wednesday 28 October 2026 at 1pm

The goal of this project is to apply symbolic and numeric integration to compute the coefficients to approximations functions in an orthogonal basis of sine and cosine functions. The project description was made via a Jupyter notebook, posted at the course web site. A good start of the project is to download and run that notebook.

1. An Orthogonal Basis of Periodic Functions

Consider $\sin(k\pi x)$ for k ranging from 1 to 5, over the interval $x \in [-1, +1]$. These 5 functions are orthogonal with respect to the inner product

$$\langle f, g \rangle = \int_{-1}^{+1} f(x)g(x)dx.$$

If we compute all inner products for those five functions, then we obtain the identity matrix, which demonstrates the orthogonality of the basis. Let us then consider $\cos(k\pi x)$ for k ranging from 0 to 5, also over the interval $x \in [-1, +1]$ and also these cosine functions define an orthogonal basis with respect to the inner product defined above. If we add the sequence of sine functions to the cosine functions, we obtain a basis of 11 functions.

For any function defined $f(x)$ defined over $x \in [-1, +1]$, we approximate f as a linear combination of cosine and sine functions running at increasing frequencies. For some positive integer n:

$$f(x) \approx \frac{1}{2}a_0 + \sum_{k=1}^{n} a_k \cos(k\pi x) + \sum_{k=1}^{n} b_k \sin(k\pi x),$$

where the coefficients are computed as

$$a_k = \int_{-1}^{+1} f(x)\cos(k\pi)dx, \quad k = 0, 1, \ldots, n,$$

and

$$b_k = \int_{-1}^{+1} f(x)\sin(k\pi)dx, \quad k = 1, 2, \ldots, n.$$

The formulas for the coefficients follow from the linearity of the integral operator and the orthogonality of the basis.

If we take $f = 2\cos(3\pi x) + 4\sin(5\pi x)$, then we can recover the coefficients in the approximations via the evaluation of the integrals.

Assignment One. Consider $f = (\sin(2\pi x))^3$.

1. Compute approximations for f for increasing values of n.
2. Verify the accuracy of the approximations A via plots of f and A over $[-1, 1]$.
3. What should the value of n be for the approximation A to match f?

For this problem, exact values for the coefficients in the approximation A are most appropriate.

2. Approximating the Exponential Function

Naturally, the basis of sine and cosine functions works very well to approximate periodic functions. But what if the function is not periodic? The next assignment considers the exponential function.

Assignment Two. Consider $f = \exp(x)$.

1. Compute approximations A of f for $n = 5, 10, 20$. For each approximation A, plot f and A. Plot also the error $\text{abs}(f - A)$ for $x \in [-1, +1]$.
2. Describe the shape of the approximation and the error, comparing the accuracy of the approximations for increasing values of n.

3. Approximation of a Piecewise Function

Often, a symbolic antiderivative may not exist, or even as it exists, it may not be useful, as is the case of a piecewise function. For the next assignment, consider the following:

$$f(x) = \begin{cases} -x - 1 & \text{if } x \in [-1, -1/2], \\ x & \text{if } x \in (-1/2, +1/2), \\ -x + 1 & \text{if } x \in [+1/2, +1]. \end{cases}$$

Although we can compute a symbolic antiderivative of a piecewise function, as this antiderivative is discontinuous, the symbolic definite integration fails. Therefore, we apply numerical integration as available in the `quad` function of the `integrate` module of `scipy`.

The input to `quad` is a Python function. The piecewise function can be turned into a Python function via substitution. Observe that `quad` returns a tuple. The first element in the tuple is the approximation for the integral, the second element in the tuple is an estimate for the error. In defining the basis functions, it is important to use the sine and cosine function of `numpy` and also the approximation for π of `numpy`.

Assignment Three. Consider the piecewise function f as defined above.

1. Apply numerical integration to compute the coefficients for an approximation A for f, for $n = 5$. Plot the function f and the approximation A. Also make a plot of the error $\text{abs}(f - A)$.
2. Describe the shape of the approximation A and the error. Where is the error largest?

4. the Deadline is Wednesday 28 October 2026 at 1pm

Upload your solution to gradescope before 1pm on Wednesday 28 October 2026.

The solution consists of one single notebook, organized according to the assignments. Mark the start of each solution to an assignment using a heading in a cell. Your notebook should run from top to bottom as a program without errors. Apply proper formatting in your notebook so it reads like a technical report if you would print it. Document the execution cells with complete sentences, properly formatted in markdown cells.

You may (not must) work in pairs for this project. A pair consists of two, not three or more. If you decide to work in a pair, then you must send an email with the name of your partner and with the email address of your partner in the copy of the email to `janv@uic.edu`, before 5pm on Wednesday 21 October. If working in a pair, then only one Jupyter notebook should be submitted.

If you have questions or difficulties with the project, feel free to ask questions during office hour.